Well-foundedness of an ordinal carries over to its natural product with any finite coefficient

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Editorial note. The counted unit is the article's own Lemma lem:wf-product at source lines 411-413 together with the proof environment that begins on the line after the statement ends (415-423); the pairing is the ordinary adjacent one, and the proof contains the author's own commented-out alternative argument at source lines 418-422, which is retained exactly as the author left it.

Editorial note. The retained context at source line 390 is the author's own announcement of the two combinatorial lemmas and of the natural (Hessenberg) operations; the retained context at source lines 392-398 is his own definition of the natural sum and of alpha ntimes k; the retained context at source line 400 is his own comparison of the natural product with the usual ordinal product, which the proof uses.

Editorial note. The retained preamble reproduces verbatim the author's document class at source line 2, his theorem declarations at source lines 16-25, his macro definitions at source lines 46-88 and 91, his natural-operation macros at source lines 104-107 and his quotation macro at source line 116. The LS2 font-encoding block at source lines 93-99, which declares \smashtimes through the stix fonts, is not retained because the host TeX Live has no stix directory and because the author never uses that symbol: source line 105 defines \ntimes as \dottimes and source line 110 shows the alternative already commented out. mathabx (source line 103) is retained because \dotplus and \dottimes come from it. The cleveref load at source line 6 and the inputenc load with the utf8x option at source line 8 are not retained, for the assembler's own cleveref machinery and the forced xelatex engine respectively. Nothing was renamed, redefined or added.

Editorial note. No retained line contains a citation or a cross-reference, so the material carries no bibliography and needs no navigation pointer.

Editorial note. The article's conservation theorems, its lemma lem:shore-blocking-wf, its lemmas lem:coh-wf-forcing-jump and lem:coh-wf-forcing-wf, and its proposition prop:coh-wf are not counted here and are not used as premises.

Allowed standard prerequisites (editorial).

- Natural sum and natural product of ordinals below epsilon_0. For ordinals alpha and beta below epsilon_0 written with a common family of exponents, the natural sum is obtained by adding the coefficients of like powers of omega, and alpha ntimes k is the natural sum of alpha with itself k times; the article's own definition is retained at source lines 392-398, together with the comparison alpha ntimes k at most alpha times omega at source line 400.. The counted lemma concludes WF(alpha ntimes k) from WF(alpha), so both the natural product and its comparison with the usual product must be available.

- Theories RCA_0 and the well-foundedness statement WF. RCA_0 is the standard base theory of reverse mathematics, and WF(alpha) states that the ordinal alpha is well founded; a model M satisfies WF(alpha) when it has no M-infinite descending sequence of ordinals below alpha; the article uses these notations throughout, for instance when it defines the Pi^1_1 proof-theoretic ordinal in its introduction.. The counted lemma is stated for a model M of RCA_0 plus WF(alpha) and concludes WF(alpha ntimes k) inside the same model.

Before working on Π^1_1-conservation of WKL and COH, we prove two combinatorial lemma which say that Shore blocking comes for free for preserving $\mathsf{WF}(\alpha)$ for any ordinal $\alpha \leqslant \epsilon_0$. Instead of using the standard sums and products over ordinals, the natural (Hessenberg) sums and products will be more convenient:

Definition 0.1 (Natural sum and product)**.** *Let α and β be two ordinals less than ϵ_0. Let $\alpha = \omega^{\gamma_1} n_1 + \cdots + \omega^{\gamma_k} n_k$ and $\beta = \omega^{\gamma_1} m_1 + \cdots + \omega^{\gamma_k} m_k$ (We allow the n_i and m_i to be equal to 0 in order to write α and β using the same exponents γ_i). We define the natural sum of α and β (denoted by $\alpha \dot{+} \beta$) to be the following ordinal (It does not depend on the choice of the family (γ_i)) :*

$$\omega^{\gamma_1}(n_1 + m_1) + \cdots + \omega^{\gamma_k}(n_k + m_k)$$

We also define $\alpha \dot{\times} k$ to be equal to be the natural sum of α with itself k times.

Note that $\alpha \dot{\times} k \leqslant \alpha \times \omega$, where $\alpha \times \omega = \sup_{k<\omega} \alpha \times k = \omega^{\gamma_1 + 1} n_1$ is the usual ordinal product.

Lemma 0.2. *lemma]lem:wf-product Let $\mathcal{M} = (M, S) \models \mathsf{RCA}_0 + \mathsf{WF}(\alpha)$ for α an ordinal less than ϵ_0. Then $\mathcal{M} \models \mathsf{WF}(\alpha \dot{\times} k)$ for every $k \in M$.*

Proof. Since $\alpha \dot{\times} k \leqslant \alpha \times \omega$ for every $k \in M$ and RCA_0 proves that the product of two well-orders is a well-order, hence $\mathsf{RCA}_0 \vdash \mathsf{WF}(\alpha) \to \mathsf{WF}(\alpha \times \omega)$, we get the desired result.

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